Stefano paggi

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Research

University of Toulouse, Toulouse, France *2023-2026*

Ph.D., Physics *(in process of completion)*

I focused on the implementation and application of the three-particle, one-electron removal/addition multichannel Dyson equation ((3,1)-MCDE), developed in the research group of Dr. Pina Romaniello and Dr. Arjan Berger. The method uses three-body Green's functions and the Dyson equation to calculate excitation and photoelectron spectra with high accuracy.

- **Scientific Software Development**: Developed a high-performance Python and FORTRAN framework for simulating electron excitation spectra using advanced many-body methods (Green's functions, Dyson equations)
- **Large-Scale Numerical Methods**: Engineered solutions for extreme-scale linear algebra problems (up to 10^{11} matrix elements) using sparse representations and general subspace reduction
- **Performance Optimization**: Optimized matrix construction and diagonalization pipelines, achieving significant performance gains in large-scale simulations
- **High Performance computing**: Deployed and executed workloads on High Performance Computing (HPC) systems, enabling scalable simulations of multi-electron systems
- **Numerical Stability & Accuracy**: Implemented numerically stable algorithms for extracting correlated excitation spectra, ensuring high accuracy for quantum systems

Education

Heidelberg University, Heidelberg, Germany *2016-2023*

M.Sc., Condensed Matter Physics, 2023 (GPA: 3.8/4.0 (German scale: 1.2, High Distinction)) *2020-2023*

- Applied the Quanty package using Lua scripting and Wolfram Mathematica to study correlated electron excitations
- Modeled a cerium impurity in a silicon environment using a single impurity Anderson model (SIAM)
- Simulated temperature-dependent spectra via Boltzmann statistics and compared with experimental observations
- Analyzed asymmetric *f*-*f* excitation features, identifying weaker-than-expected temperature dependence
- Focus: physical modeling, simulation workflows, and interpretation of computational results

B.Sc., Physics, 2016 (German scale: 2.5, Good) *2016-2020*

- Collaborated within an experimental physics group on the modeling and analysis of a ^{3}He atomic beam spectrometer
- Computed beam trajectories and scattering cross-sections, supporting the design and interpretation of experiments
- Developed tools for wavelet scattering representations for signal characterization

Relevant Coursework: Linear Algebra, Scientific Computing, Quantum Statistics, Numerical Methods, Advanced Mathematics for Physicists

Skills

- **Computing:** Python (NumPy, SciPy, PySCF), Wolfram Mathematica, Lua, Java, FORTRAN, Git, emacs, LaTeX, Linux/Shell, High Performance Computing (Slurm)
- **Languages:** English (*full professional*), German *(native)*, Italian *(native)*, French *(elementary)*

Publications

- *Ground and excited-state properties of the extended Hubbard dimer from the multichannel Dyson equation.* S. Paggi, J.A. Berger, P. Romaniello. Journal of Chemical Physics, **163**, 154109 (2025).

  I tested how the (3,1)-MCDE performs for the exactly solvable extended Hubbard dimer, which is an H_2-molecule-like model. The photoemission spectra are compared to the GW, Second Born, and exact spectra. I also calculated the Galitskii–Migdal potential energy surface. The (3,1)-MCDE performed better than the other theories considered.

- *Multichannel Dyson equations for even- and odd-order Green's functions: Application to double excitations.* G. Riva, T. Fischer, S. Paggi, J.A. Berger, P. Romaniello. Physics Review B **111**, 195133 (2025).

  I applied the (4,0)-MCDE (double neutral excitations) to a helium-like two-level model. I compared the calculation to a state-of-the-art GW calculation, which creates an unphysical peak for the double excitation, unlike our method. I also analyzed the (4,0)-MCDE for different compositions of its effective Hamiltonian.

- *Core and valence photoemission spectra of atoms and molecules from a multichannel Dyson equation.* S. Paggi, J.A. Berger, P. Romaniello. *Submitted to JCP. arXiv:2607.20070* (2026).

  I implemented the (3,1)-MCDE in a Python code that works with the PySCF package to calculate molecules such as NH_3, BeH_2, LiH, FH, H_2O, and the noble gas neon. The code is generally applicable to any many-body physics code that includes two-electron integrals. It is also able to return the full spectra, as well as the character contributions of each satellite peak. A Haydock–Lanczos solver is also included in the implementation.

- *Benchmark of Multi-Channel Dyson Equation and Algebraic Diagrammatic Construction Methods for molecules.* M. Keizer, S. Paggi, J. A. Berger, P. Romaniello, A. Förster *arXiv:2608.25669* (2026).

  I provided the (3,1)-MCDE implementation in FORTRAN for Keizer and Förster. Starting from this code, they were able to perform all the (3,1)-MCDE calculations.

Conferences

- APS March Meeting 2026, Denver, Colorado. *March 2026*
- Materials Science from First Principles 2: HANAMI-CECAM Materials Science, Paris, France. (*poster*) *November 2025*
- Interdisciplinarity conference on many-body theory, Nancy, France. *June 2025*
- Green's function methods: the next generation 6, Toulouse, France. *October 2024*
- 4th edition of the International summer School in electronic structure Theory: electron correlation in Physics and Chemistry (ISTPC 2024), Aussois, France. (*poster*) *June 2024*
- 20th ETSF Young Researchers' Meeting (Toulouse, 27th - 31st May 2024), Toulouse, France. *May 2024*